

Javad Komijani

Senior Computational Physicist

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Professional Summary

Senior Computational Physicist with over 10 years of experience in scientific computing and 5 years in scientific machine learning. Develops generative models (diffusion models, normalizing flows) in PyTorch for large-scale simulations. Expert in differential equations, Bayesian inference, and numerical optimization. Bridges advanced mathematical modeling with machine learning to solve complex computational problems.

Technical Skills

Programming Python, C/C++, Cython, Bash
ML PyTorch, Generative Models, Diffusion Models, Scientific ML, Distributed Training
Applied Math Differential equations, Bayesian inference, Monte Carlo, Optimization
Systems Linux, HPC clusters, Parallel computing, Git
Data NumPy, SciPy, Pandas, Scikit-learn, SQL

Professional Experience

2021–Present **Senior Postdoctoral Researcher, ETH Zurich**

- Designed and implemented GPU-based generative models for large-scale lattice simulations.
- Developed custom PyTorch extensions for differentiable linear algebra and ODE solvers.
- Executed computational workflows on HPC clusters.
- Contributed ML modules to the interTwin Digital Twin Engine; supervised MSc/PhD students.

2015–2021 **Postdoctoral Researcher, TU Munich / Glasgow / Tehran**

- Applied Bayesian inference and Monte Carlo simulations to large-scale numerical datasets.
- Developed scalable lattice-QCD simulation pipelines in HPC environments.
- Collaborated internationally in computational and mathematical physics projects.

Selected ML & Scientific Computing Projects

Normflow Normalizing Flows in PyTorch; supporting complex high-dimensional scientific distributions.

Diffusion Diffusion-based generative models for structured scientific data on Lie group manifolds.

Linear Alg. Linear Algebra Extension for PyTorch; supporting automatic differentiation for eigenvalue and singular value decompositions of unitary matrices.

Differentiable ODE Solver Adjoint-based ODE solver extension supporting automatic differentiation for scientific ML workflows.

Education

2015 **PhD in Physics, Washington University in St. Louis**
2009 **MSc in Electrical Engineering, University of Tehran**

Publications

Selected peer-reviewed publications in computational physics and scientific ML. Full list available upon request.

Languages

English Fluent
German Elementary